

Администрирование локальных сетей

Лабораторная работа №8

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Цели и задачи работы

Приобретение навыков настройки сетевых сервисов DHCP и DNS в локальной сети.

Выполнение лабораторной работы

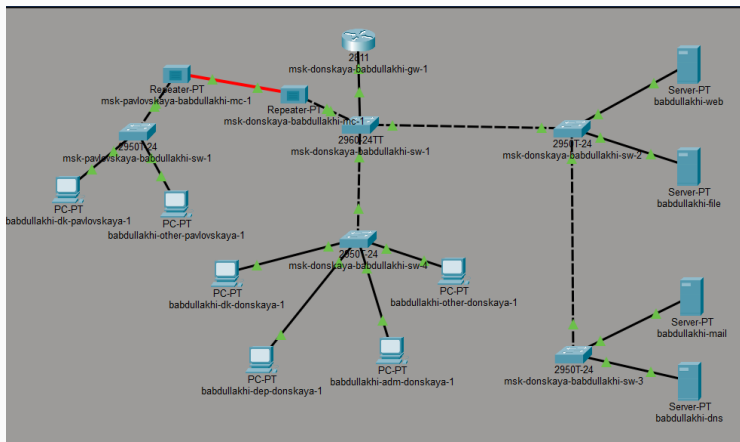


Рис. 1: Топология сети

Настройка DNS-сервера

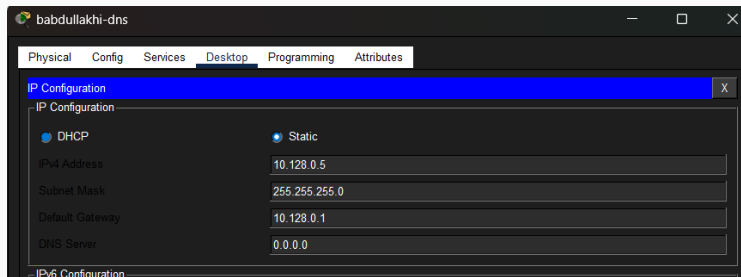


Рис. 2: Настройка IP-адреса DNS-сервера

Настройка DNS-сервиса

babdullakhi-dns

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

_____ **A Record**

Add Save Remove

No.	Name	Type	Detail
0	dns.donskaya.rudn.ru	A Record	10.128.0.5
1	file.donskaya.rudn.ru	A Record	10.128.0.3
2	mail.donskaya.rudn.ru	A Record	10.128.0.4
3	www.donskaya.rudn.ru	A Record	10.128.0.2

Настройка DHCP на маршрутизаторе

```
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#ip name-server 10.128.0.5
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#service dhcp
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp pool dk
msk-donskaya-babdullakhi-gw-1(dhcp-config)#network 10.128.3.0 255.255.255.0
msk-donskaya-babdullakhi-gw-1(dhcp-config)#default-router 10.128.3.1
msk-donskaya-babdullakhi-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-babdullakhi-gw-1(dhcp-config)#exit
^
% Invalid input detected at '^' marker.

msk-donskaya-babdullakhi-gw-1(dhcp-config)#exit
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp excl
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.3.1 10.128.3.29
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.3.200 10.128.3.254
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp pool departaments
msk-donskaya-babdullakhi-gw-1(dhcp-config)#network 10.128.4.0 255.255.255.0
msk-donskaya-babdullakhi-gw-1(dhcp-config)#default-router 10.128.4.1
msk-donskaya-babdullakhi-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-babdullakhi-gw-1(dhcp-config)#exit
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.4.1 10.128.4.29
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.4.200 10.128.4.254
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp pool adm
msk-donskaya-babdullakhi-gw-1(dhcp-config)#network 10.128.5.0 255.255.255.0
msk-donskaya-babdullakhi-gw-1(dhcp-config)#default-router 10.128.5.1
msk-donskaya-babdullakhi-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-babdullakhi-gw-1(dhcp-config)#exit
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.5.1 10.128.5.29
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.5.200 10.128.5.254
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp pool other
msk-donskaya-babdullakhi-gw-1(dhcp-config)#network 10.128.6.0 255.255.255.0
msk-donskaya-babdullakhi-gw-1(dhcp-config)#default-router 10.128.6.1
msk-donskaya-babdullakhi-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-babdullakhi-gw-1(dhcp-config)#exit
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.6.1 10.128.6.29
msk-donskaya-babdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.6.200 10.128.6.254
msk-donskaya-babdullakhi-gw-1(config)#exit
msk-donskaya-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```


Получение IP по DHCP

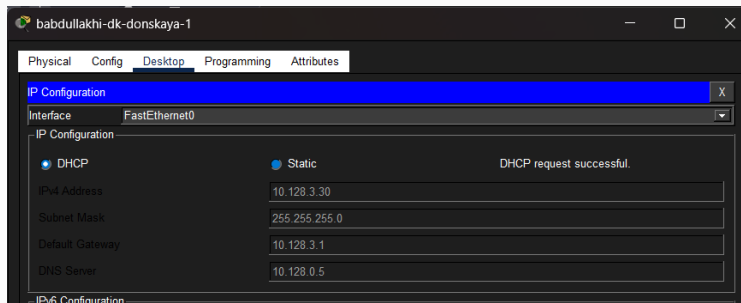


Рис. 5: DHCP на ПК

Проверка соединения

```
babdullakhi-dk-donskaya-1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.3.31

Pinging 10.128.3.31 with 32 bytes of data:

Reply from 10.128.3.31: bytes=32 time<1ms TTL=128
Reply from 10.128.3.31: bytes=32 time<1ms TTL=128
Reply from 10.128.3.31: bytes=32 time<1ms TTL=128
Reply from 10.128.3.31: bytes=32 time<1ms TTL=128

Ping statistics for 10.128.3.31:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.4.30

Pinging 10.128.4.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.6.30

Pinging 10.128.6.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.6.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

```
Pool dx :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 2
Excluded addresses : 8
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.128.3.1 10.128.3.1 - 10.128.3.254 2 / 8 / 254

Pool departaments :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Excluded addresses : 8
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.128.4.1 10.128.4.1 - 10.128.4.254 1 / 8 / 254

Pool adm :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Excluded addresses : 8
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.128.5.1 10.128.5.1 - 10.128.5.254 1 / 8 / 254

Pool other :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 2
Excluded addresses : 8
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.128.6.1 10.128.6.1 - 10.128.6.254 2 / 8 / 254
mek-donakava-bahdullakhi-gw-14
```

```
mek-donskaya-babdullakhi-gw-1#  
mek-donskaya-babdullakhi-gw-1#sh ip dhcp bin
```

IP address	Client-ID/ Hardware address	Lease expiration	Type
10.128.3.30	00E0.B097.AEC8	--	Automatic
10.128.3.31	0002.1705.C226	--	Automatic
10.128.4.30	0007.EC9C.A704	--	Automatic
10.128.5.30	00E0.A37A.D875	--	Automatic
10.128.6.30	00D0.97E7.E1C7	--	Automatic
10.128.6.31	0006.2AA0.9A8B	--	Automatic

```
mek-donskaya-babdullakhi-gw-1#
```

Рис. 8: DHCP binding

DHCP Discover

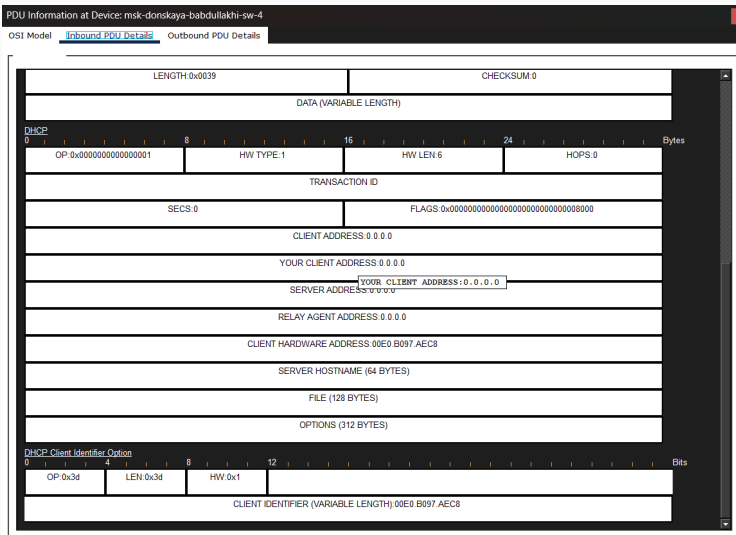


Рис. 9: DHCP Discover

DHCP Offer

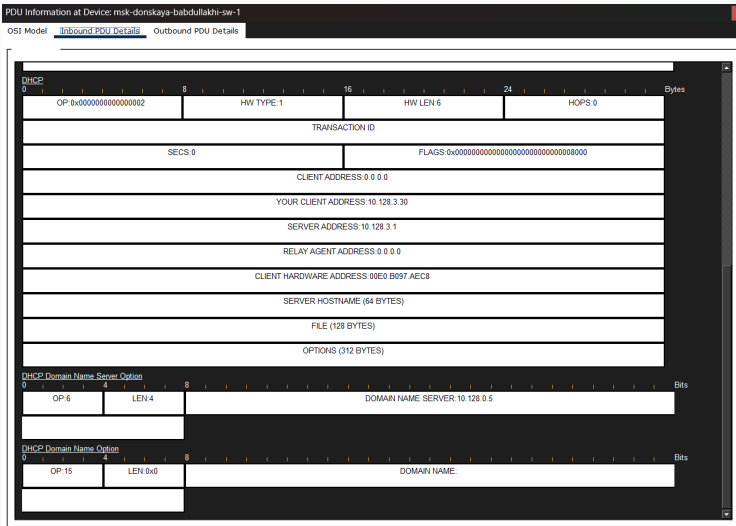
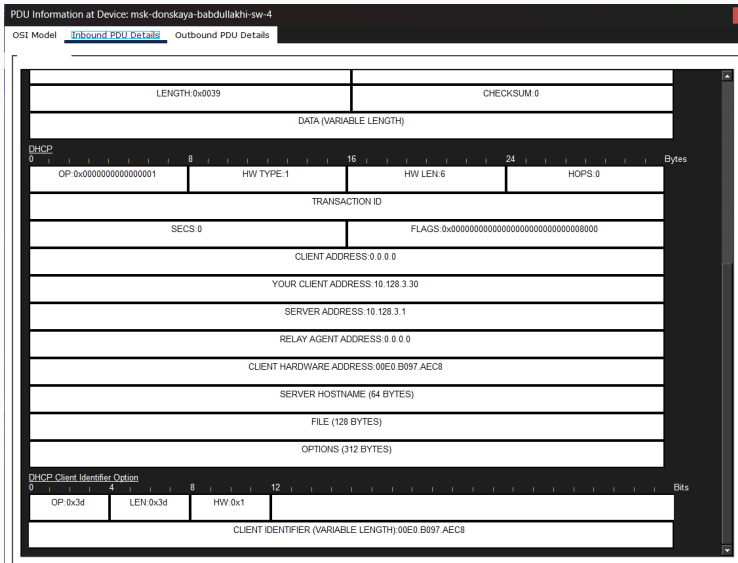


Рис. 10: DHCP Offer

DHCP Request



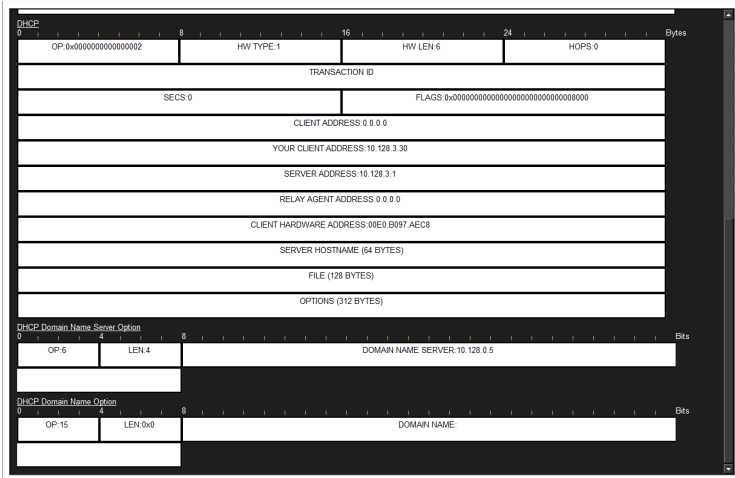


Рис. 12: DHCP ACK

Процесс DHCP в симуляции

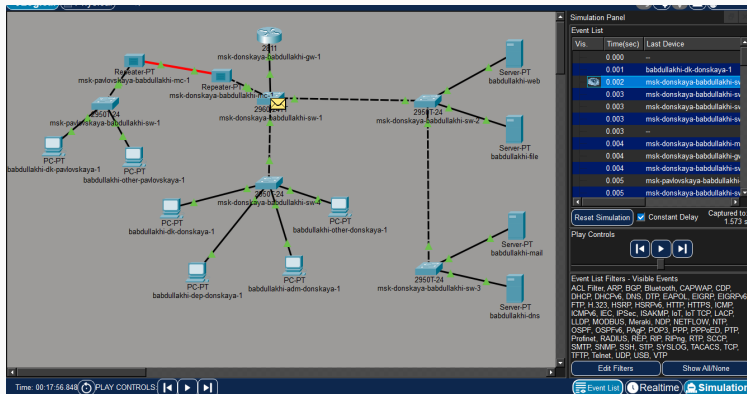


Рис. 13: Simulation

Выводы по проделанной работе

В ходе лабораторной работы была выполнена настройка DNS- и DHCP-сервисов. Настроен DNS-сервер с доменными записями, а также DHCP-сервер на маршрутизаторе с несколькими пулами адресов. Оконечные устройства успешно получают IP-адреса автоматически. Проверка связности и анализ DHCP-пакетов в режиме симуляции подтвердили корректную работу сети и настроенных сервисов.